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Intelligent Tender Management and Contract Lifecycle Platform

AI-Assisted Bid Evaluation & Monitoring System

Industry Project:2025-26
Review-I

Dept. of Computer Engineering

Group No 22

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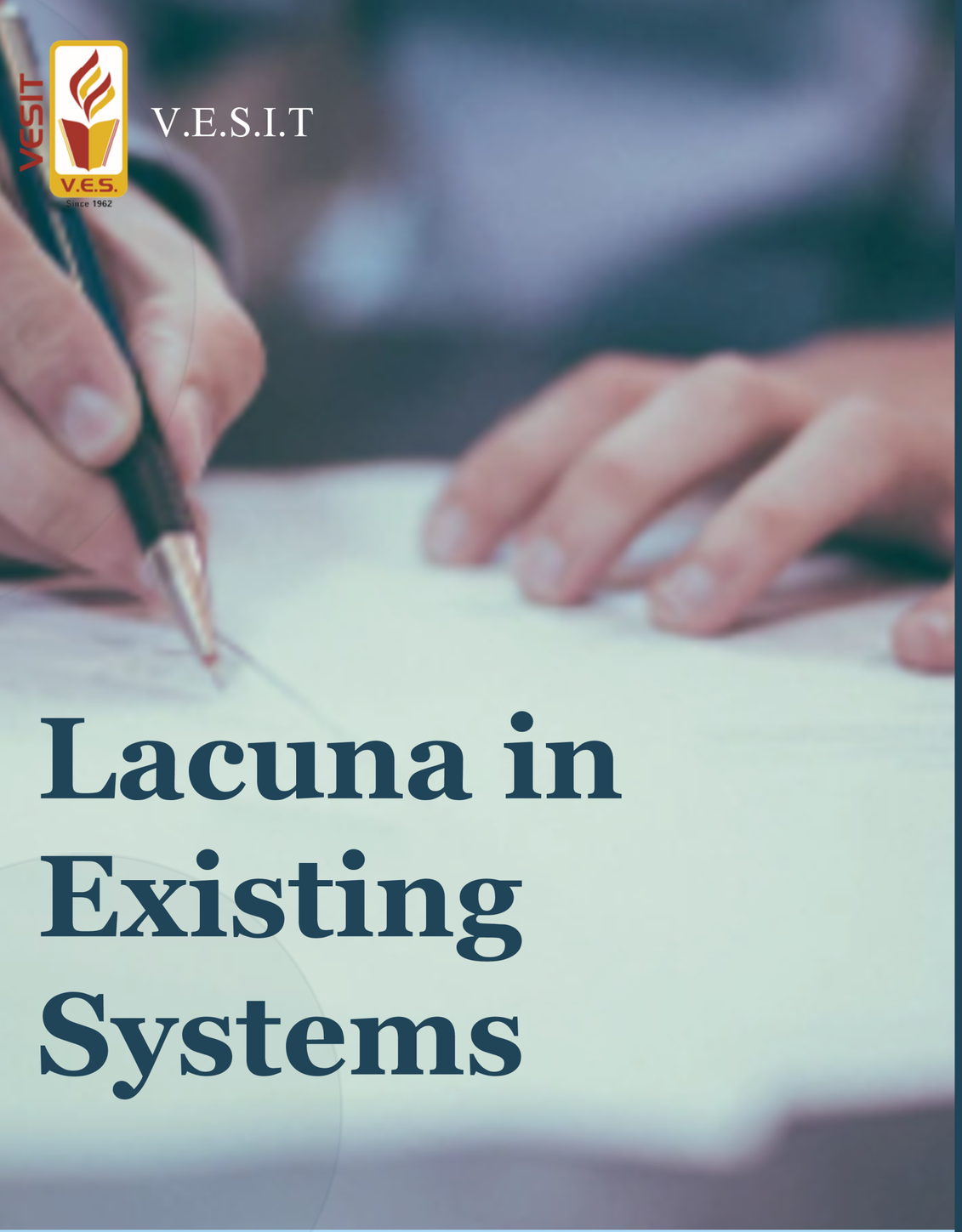
Introduction to Project

- Traditional tender management systems are manual, time-consuming, and vulnerable to errors and fraud.
- There is a need for a transparent, efficient, and technology-driven procurement system to improve accountability.
- The proposed solution is an AI-powered Tender Management and Contract Lifecycle Platform.
- The system enables intelligent bid evaluation, anomaly detection, and historical bidder profiling.
- It also supports post-award monitoring while ensuring final decisions remain under human authority.



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Lacuna in Existing Systems

Traditional manual tendering processes suffer from critical inefficiencies that compromise integrity and speed.

Manual Evaluation Delays

Evaluating complex bids is time-consuming and error-prone.

Limited Transparency & Fraud Risks

Manual systems fail to effectively detect fraud and suspicious bidding practices.

Inefficient Post-Award Tracking

Paper-based monitoring leads to inconsistent tracking of deliveries and milestones.

Lack of Historical Data Usage

Past bidder performance is rarely considered in future evaluations.

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Problem Definition

Key Challenges

Complex Bid Evaluation

Manual multi-parameter bid scoring is slow and prone to subjective errors.

Fraudulent Bids

Manual detection of shell companies and abnormal bids is difficult and unreliable.

Post-Award Monitoring

Inconsistent, paper-based tracking of delivery schedules and work milestones leads to project delays.

Lack of Historical Data

Past vendor performance, penalties, and compliance records are rarely integrated into future bid evaluations.

Project Definition

Problem Statement

Current tender management systems are manual, slow, and prone to errors and fraud. Evaluating bids, detecting suspicious activities like shell companies or duplicate bids, and tracking post-award milestones are all done without any intelligent assistance, leading to delays, poor accountability, and biased decisions. There is a need for an AI-powered platform that automates bid evaluation, detects anomalies, maintains bidder history, and monitors contracts efficiently ,while keeping final decisions in human hands.

Core Objectives:

- ✓ Transparency
- ✓ Efficiency
- ✓ Accountability

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Proposed Solution

An integrated intelligent platform comprising six core modules designed to streamline the entire tender lifecycle.

AI-Assisted Evaluation

Automated weighted scoring. Detects anomalies like shell companies, duplicate bids, and suspicious documents to assign risk scores.

Committee Review

Interface for reviewing AI-generated rankings. Allows members to accept, reject, or comment, ensuring human authority in final decisions.

Post-Award Monitoring

Checklist-based verification for supply contracts and incremental progress report uploads for work contracts.

Delay & Penalty AI

Analyzes tender clauses against actual progress reports. Calculates delay durations and recommends applicable penalties.

Bidder History

Maintains dynamic profiles with risk scores, past delays, penalties, and compliance records influencing future evaluations.

Chatbot Interface

NLP-powered assistant for querying tender status, project delays, and penalty applicability.

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Literature Survey

Research Paper	Year	Method / Technology	Relevance to Your Proposed System	Research Gap
Artificial Intelligence in E-Procurement: *A Literature Review*	2023	AI decision support, procurement analytics	Supports AI-based bid evaluation, scoring, and automated decision-making in e-tendering.	No real-time monitoring or chatbot-based decision support.
The Role of Artificial Intelligence *in the Procurement Process*	2023	Intelligent procurement lifecycle framework	Supports end-to-end procurement lifecycle management and AI-driven risk assessment.	Does not include automated delay detection or penalty recommendatic
Deep Learning for Detecting Bid Rigging	2021	Deep learning anomaly detection	Directly supports fraud detection, abnormal bid identification, and risk scoring in tender evaluation.	Not integrated with contract monitoring or decision workflow.
Artificial Intelligence for Supplier Selection *and Evaluation: A Review*	2022	AI vendor ranking and performance analytics	Supports bidder history tracking, vendor evaluation, and performance-based selection.	Does not integrate real-time monitoring or AI chatbot interface.
The Future of Procurement: Integrating AI, Automation, and Blockchain	2022	AI automation and intelligent procurement platforms	Supports integrated smart procurement platform combining analytics, automation, *and decision supp.*	Conceptual framework—lacks full practical implementation.

Table I - Literature survey

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Methodology employed

The Incremental Development Model was adopted as it allows the system to be built and delivered in functional modules.

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Hardware, Software & Tools

Hardware Environment

Cloud Infrastructure

Hosted virtual server on Render or Vercel — scalable, accessible, reliable deployment.

Workstations

Standard desktop/laptop systems for procurement officers and committee members.

Field Devices

Optional mobile device support for on-site verification and status updates from the field.

Software Stack

Frontend & Backend

React.js for responsive web UI, Node.js + Express.js for robust REST API services.

Data & AI Engine

MongoDB Atlas (NoSQL). Python microservices (NumPy, Pandas, Scikit-learn) for risk scoring.

Ai and NLP models

- Random Forest – Vendor & bid scoring
- Isolation Forest – Fraud / anomaly detection
- TF-IDF + Cosine Similarity – Document similarity detection
- spaCy NLP – Contract clause and data extraction
- RAG-based Chatbot (LLaMA) – Intelligent query handling (future work)

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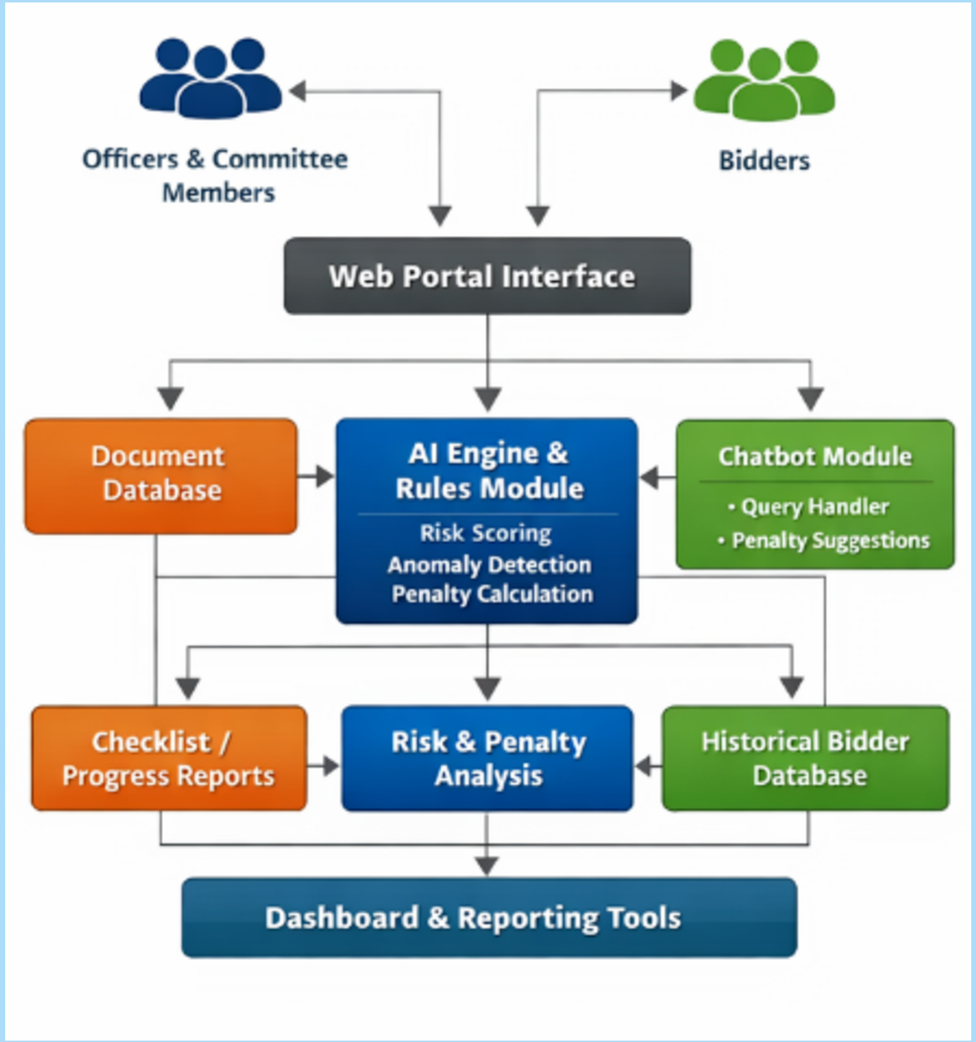

Diagram I - Block Diagram

System Architecture

The platform utilizes a scalable, microservices-based architecture separating the UI, business logic, and intelligent processing units to ensure modularity.

User Interface

React.js with responsive, role-based dashboards for Officers and Committees.

API Gateway

Node.js & Express handle auth, routing, and business logic orchestration.

AI Microservices

Python services using Scikit-learn and NLP for risk scoring and anomaly detection.

Data Persistence

MongoDB Atlas stores unstructured tender documents, user profiles, and audit logs.

Chatbot Engine

NLP-driven interface for contextual answers on delays and penalties.

Deployment

Cloud-native on Render (backend) and Vercel (frontend).

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Modular Diagram

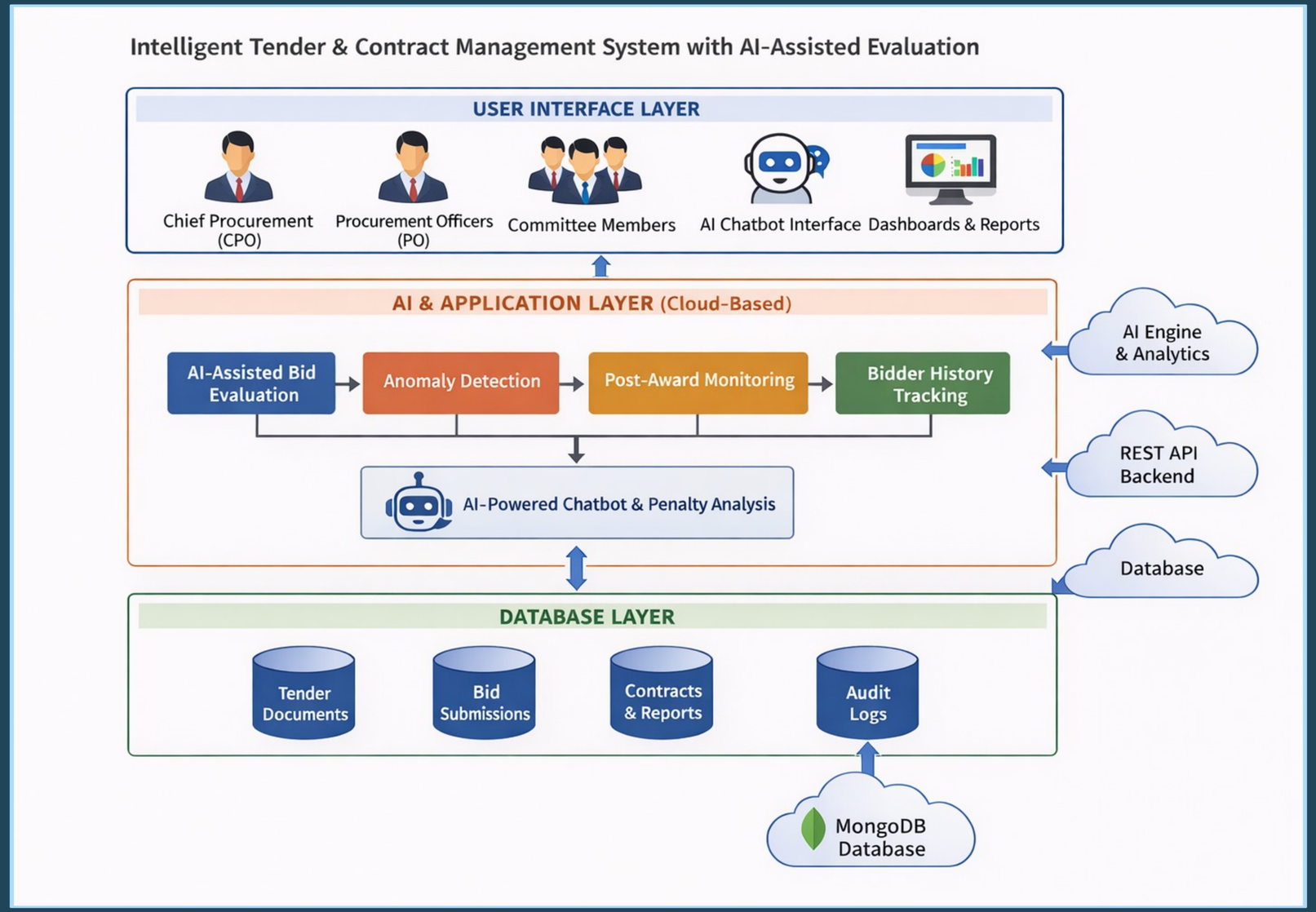
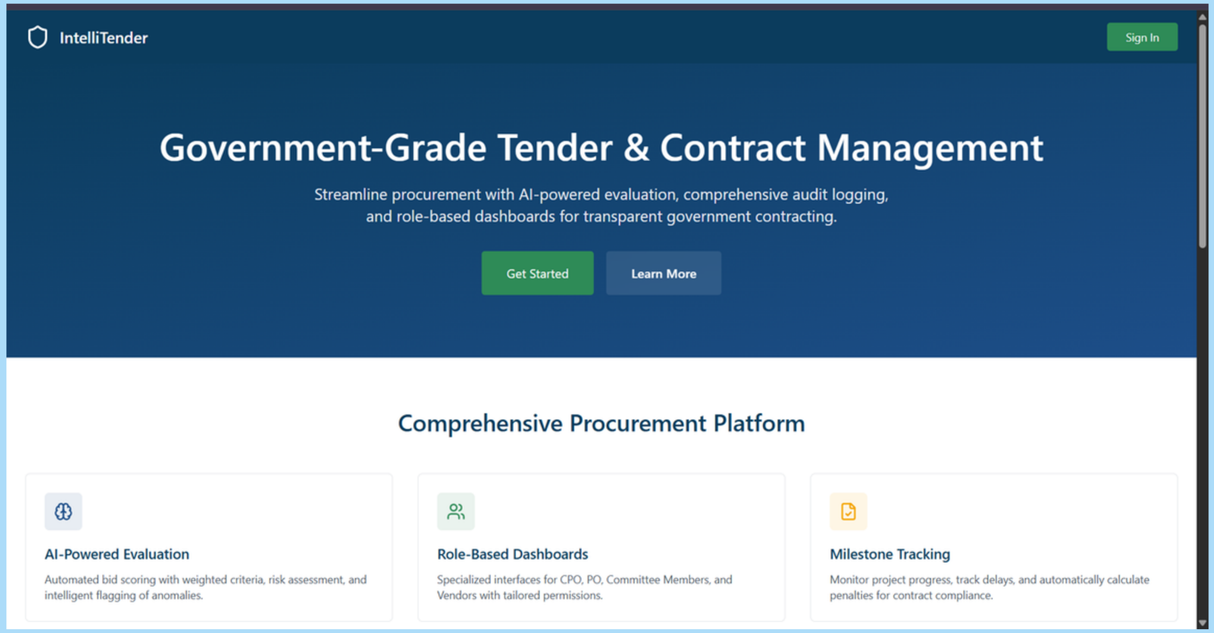
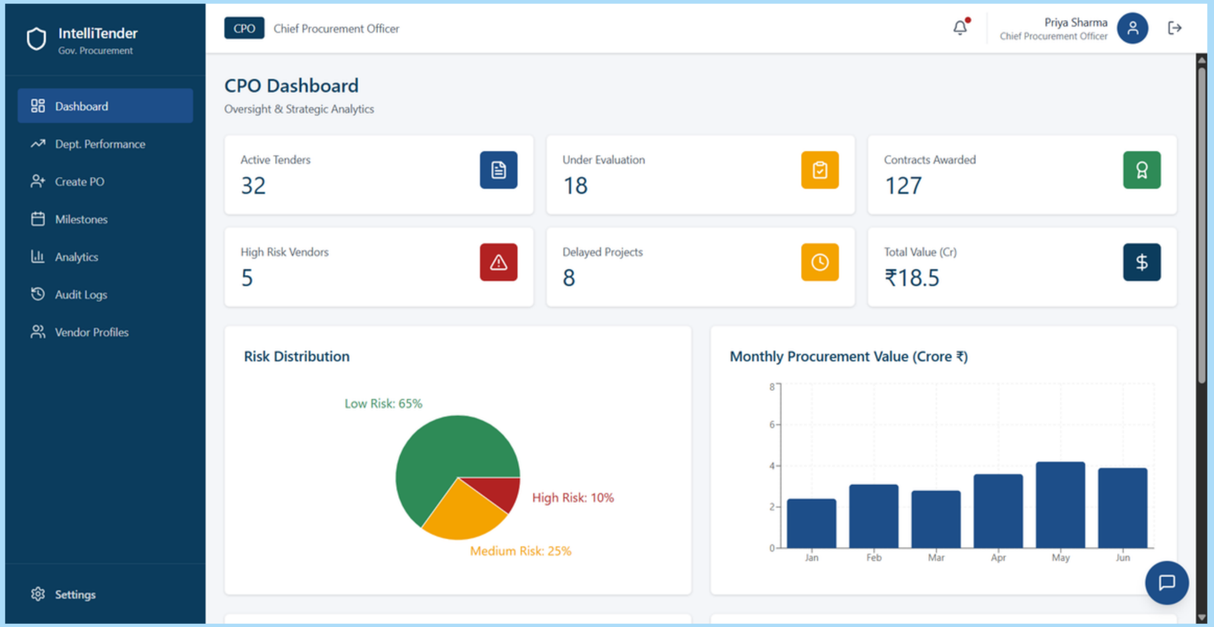
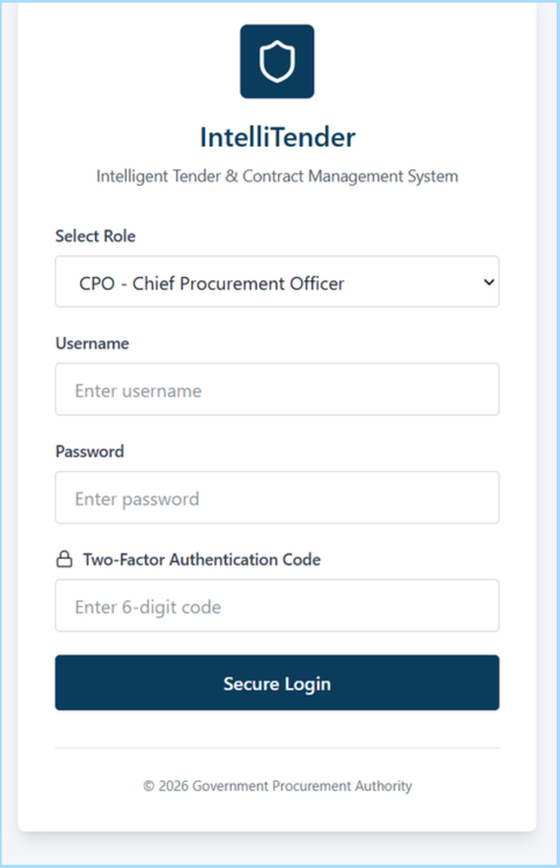


Diagram II - Modular Diagram

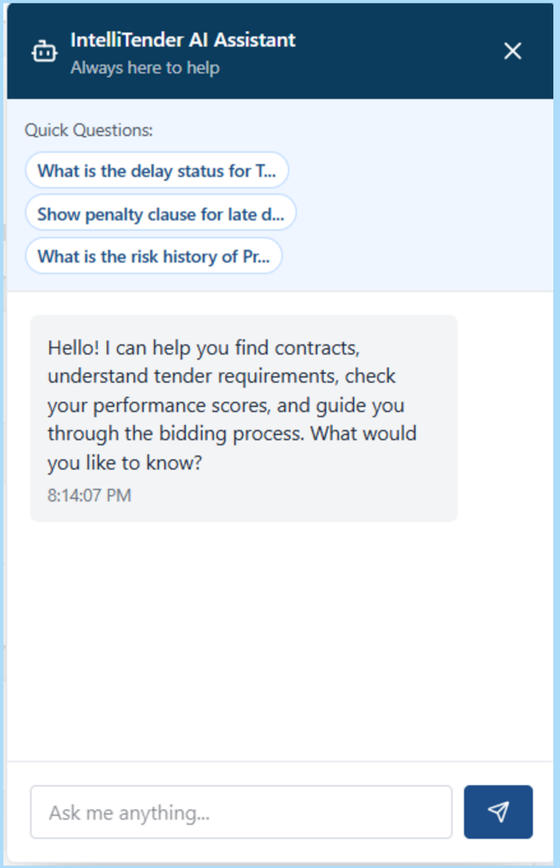
Figma / UI







[Figma Design Preview]



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Implementation (work done)

Vendor / User

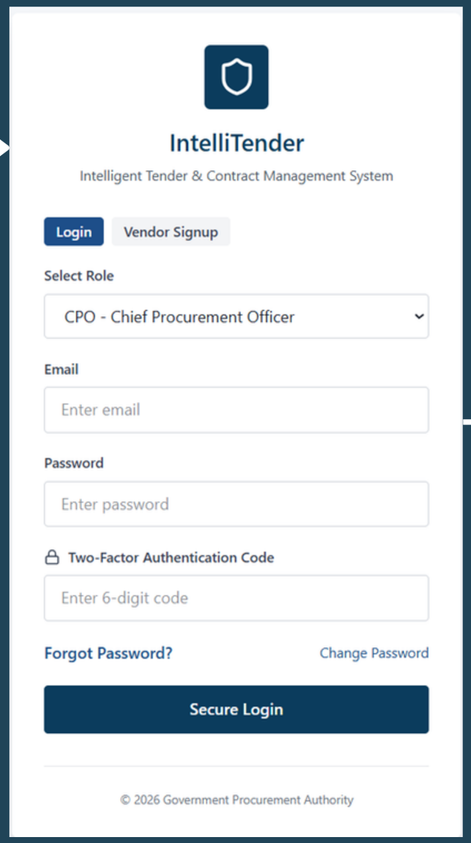
- ✓ Signup & Login
- ✓ Forgot Password
- ✓ Change Password

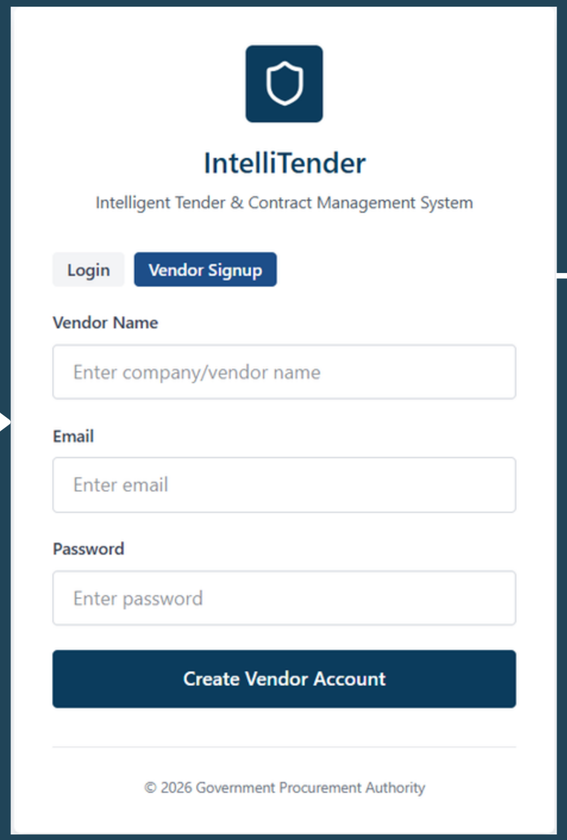
PO Officer

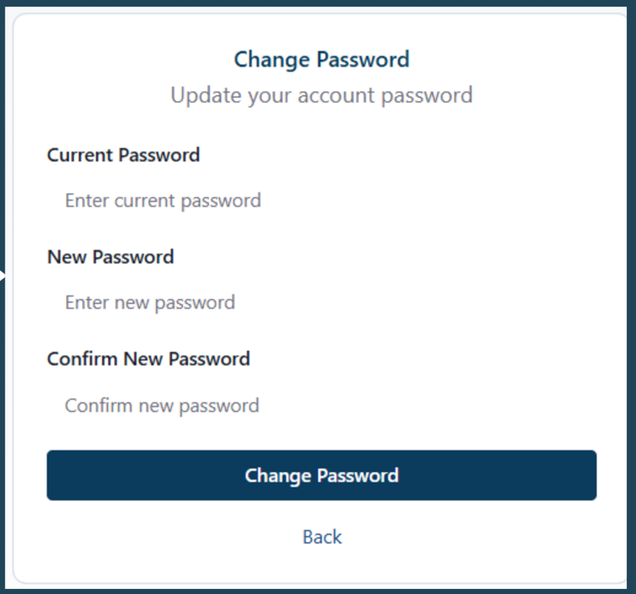
- ✓ Freeze Vendor Account
- ✓ Delete Vendor Account
- ✓ Create Committee Member Entry

CPO Officer

- ✓ Create PO Officer Entry
- ✓ Manage Officer Access







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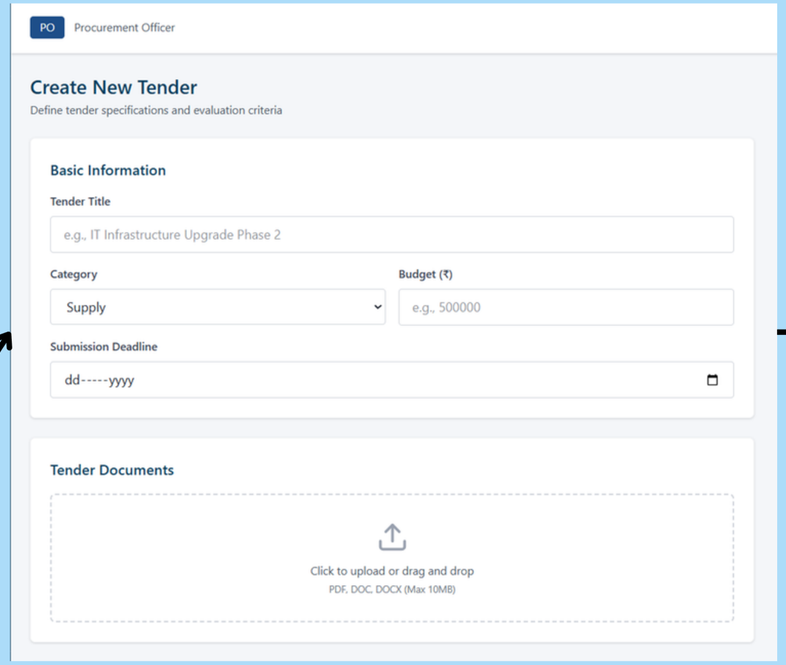

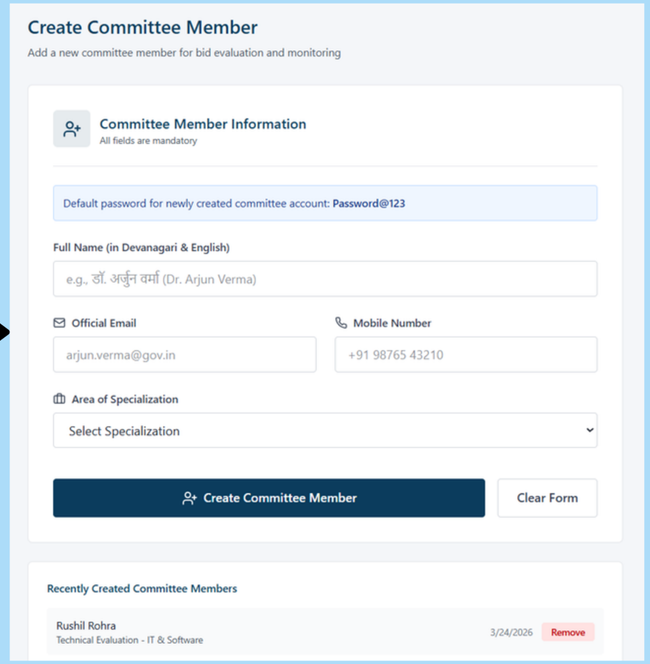

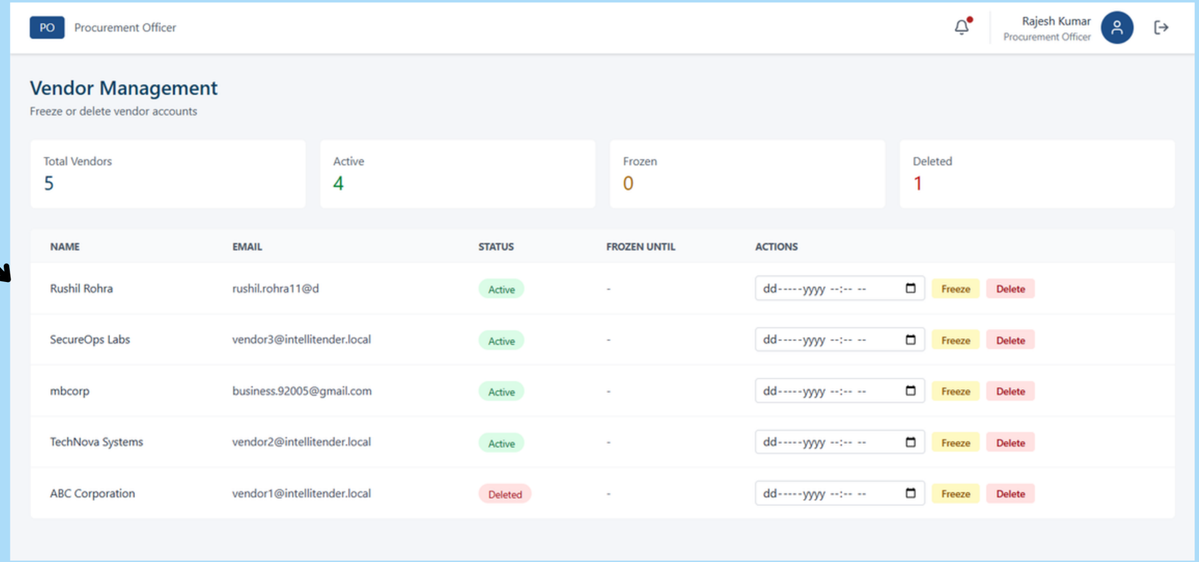

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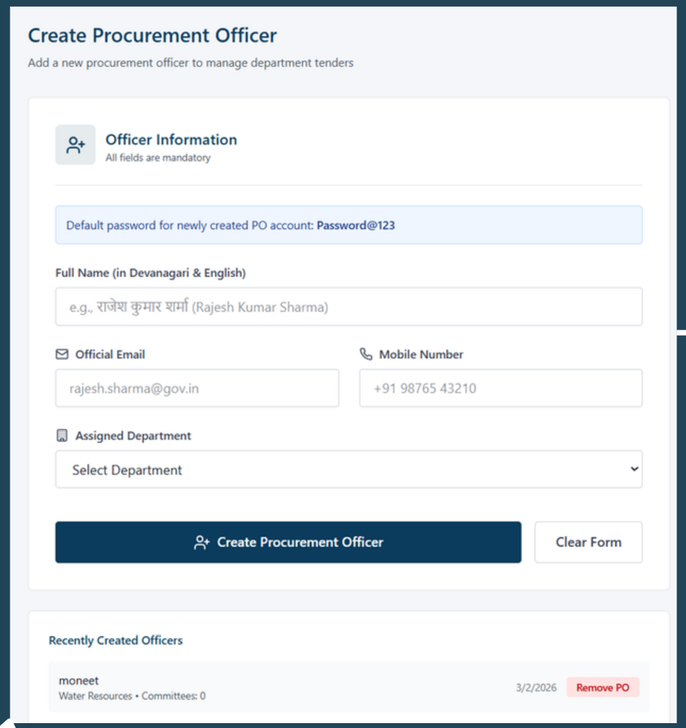

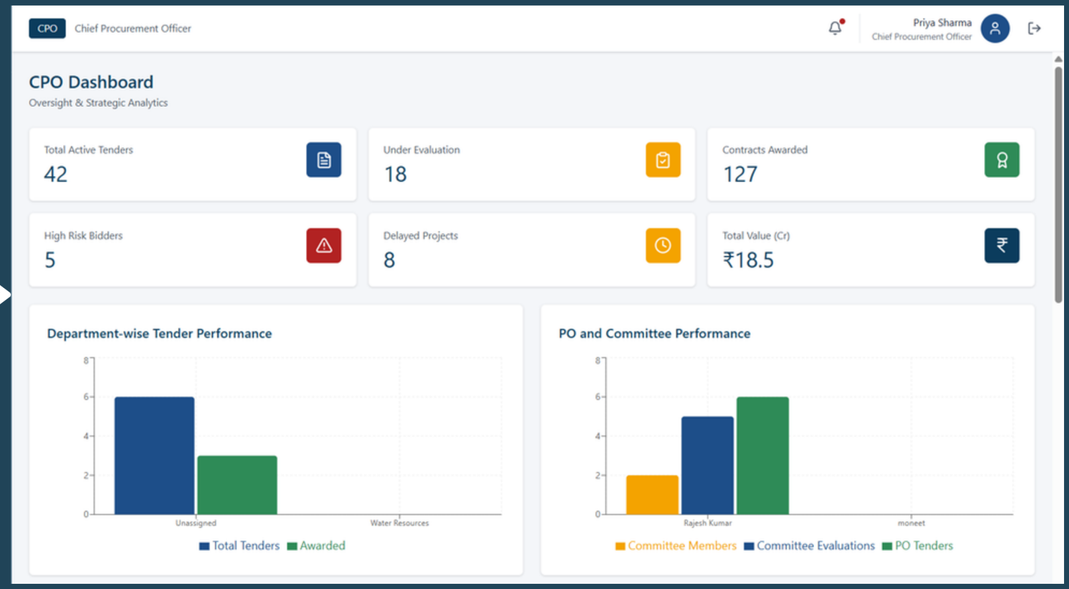

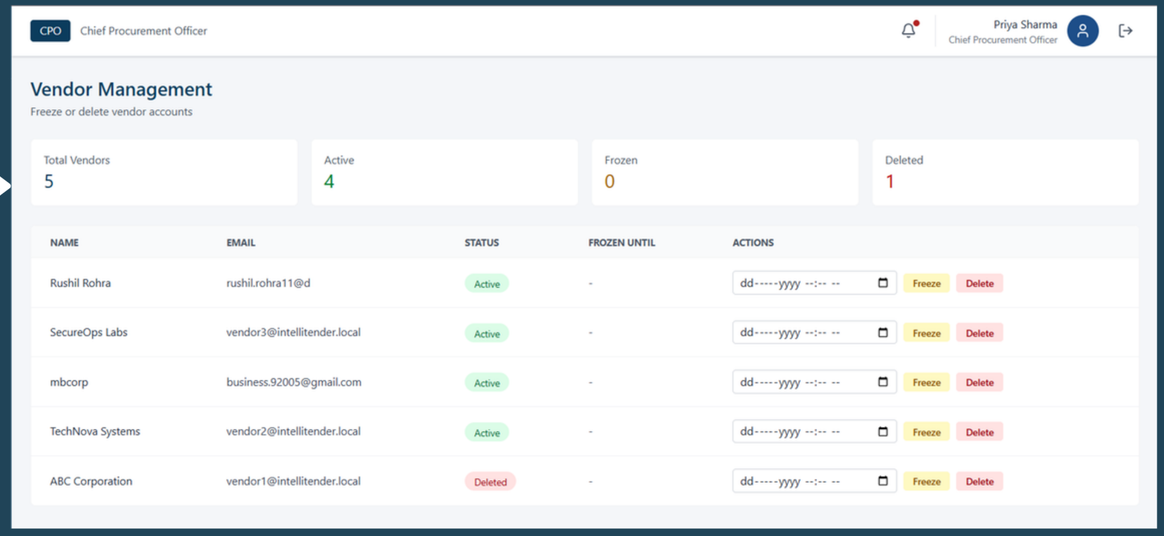

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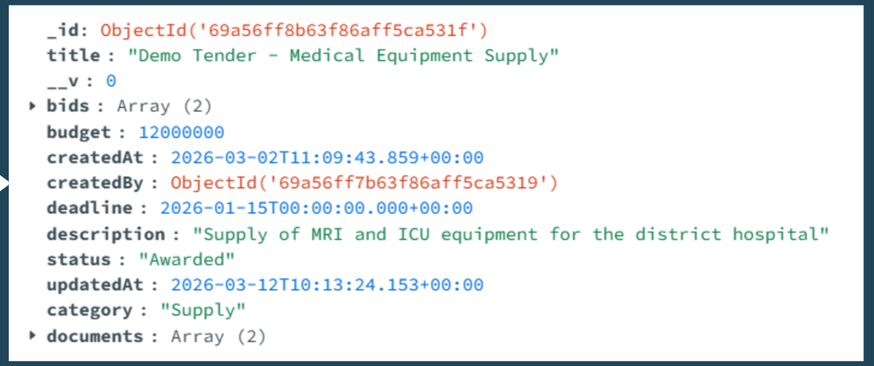
Implementation — Status Overview

✅ Completed Modules

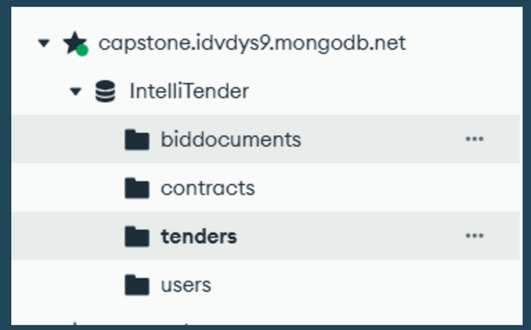
- User Authentication (Login / Signup)
- Password Management
- Role-Based Access Control
- Officer / Committee / CPO Entry

Schema Design(MongoDB) →



🚧 In Progress

- AI Bid Evaluation Module
- Bidder History & Risk Scoring
- Post-Award Monitoring
- NLP Chatbot Interface



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Next Work Plan

1. UI/UX Research
- ***Implementation Done***

2. AI Model Finalization
- Isolation Forest / Random Forest for anomaly detection & bid scoring
- RAG-based chatbot (LLaMA) for penalty & delay queries
- Finalize Python microservice architecture

Transition from theoretical design to practical implementation planning: identifying real-world constraints, refining system scope, and finalizing all technical decisions.

3. Database Schema Design (MongoDB)
- ***Partial Implementation Done***

4. API Endpoint Mapping
- Define all REST endpoints with request/response structure
- Map module interactions: Frontend → Backend → AI Service → DB
- Finalize API contracts for each module

5. Low Level Design (LLD) Document
- Module-level interaction diagrams
- Data flow for bid submission, AI evaluation, post-award monitoring
- Finalize tech decisions before implementation begins

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Conclusion

The Intelligent Tender and Contract Management System represents a significant modernization of procurement processes, preserving human decision-making authority.

AI-Assisted Modernization

Transforms manual procurement by automating complex bid evaluations and detecting anomalies. Reduces evaluation time, minimizes human error, and allows officers to focus on strategic decision-making.

Structured Monitoring & Compliance

Differentiates supply and work contracts with customized monitoring. AI-driven chatbot analyzes timelines and clauses to automatically recommend applicable penalties for delays.

Transparency & Accountability

Fully traceable digital audit trails and bidder historical profiling. Past performance directly influences future evaluations, ensuring fair, data-driven procurement outcomes.

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